Thus, we come to know, if a selection **A** can occur in ***m*** ways and another selection **B** can occur in ***n*** ways, and both the selections are **dependent** on each other then, the two selections can be performed in exactly ($m \times n$) different ways. Let as learn more about from the following examples.

Example 7.2

Praveen bought **3 shirts, 2 jeans and 3 pairs of shoes** for his birthday. In Fig. 7.5 shows the different ways of wearing the dress. In how many different ways can Praveen wear a dress on his birthday?

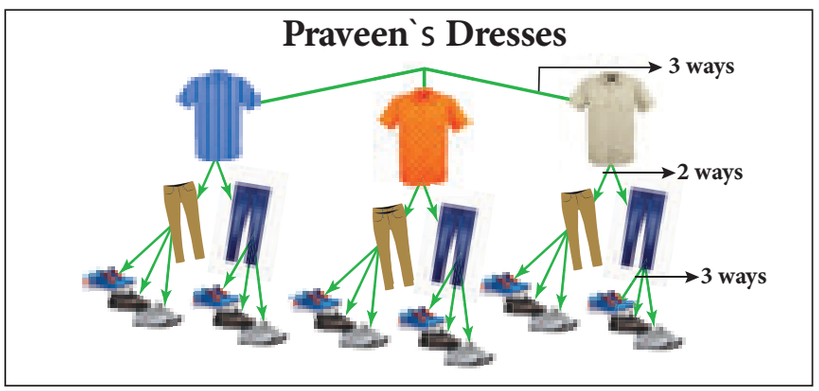


Fig. 7.5

Solution:

Here, Praveen has 3 shirts  , 2 jeans  and  3 pairs of shoes.

He can wear a dress either this way  or he can have the choices as shown in the Fig. 7.6



3 shirts × 2 jeans × 3 pairs of shoes = 18 ways

Fig. 7.6

Therefore, Praveen can wear his dress in **18** **(3×2×3)** **different possible ways** on his birthday.

Example 7.3

In class VIII, a math club has four members **M,A,T** and **H.** Find the number of different ways, the club can elect

(i) a leader,

(ii) a leader and an assistant leader.

Solution:

(i) To elect a leader

In class VIII, a math club has four members namely **M, A, T** and **H.**

Therefore, there are **4** **(4 × 1)** **different ways** by which they can be elected a leader.

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